14 April 1978
SUPERSEDING
MIL-S-55347A(EL)
11 January 1974

MILITARY SPECIFICATION

RADIO TELETYPEWRITER SET AN/GRC-142( )/ AN/GRC-122( )

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the transportable Shelter Assemblage Radio Teletypewriter AN/GRC-142( ) which provides one way reversible, radio teletypewriter, amplitude modulated voice, single side band voice and continuous wave CW communications in the high frequency band. Facilities are also provided to allow the ready installation and operation of tactical electronic teletypewriter security equipment when secure radio teletypewriter is required. Provisions are made for operation of the AN/GRC-142( ) from either 28 volts DC or 115 volts single phase 50/60 Hz AC primary power sources. Provisions are provided for installation of auxiliary equipment to permit full duplex voice and radio teletypewriter operation, as an AN/GRC-122( ), when required.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5815

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2. APPLICABLE DOCUMENTS

2.1 Issues of Documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Packaging Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For Use in Electronic, and Associated Electrical Equipment)
MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys and Titanium Alloys, Spot and Seam
MIL-T-7928	Terminal Lugs Crimp Style
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements
MIL-STD-462	Electromagnetic Interference Characteristic Measurements
MIL-STD-810	Environmental Test Methods

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DRAWINGS

ELECTRONICS COMMAND

DL-SC-B-685200	Radio Teletypewriter Set AN/GRC-142( )
DL-SC-B-685201	Shelter Electrical Equipment AN/GRC-142( )
SC-C-4500	Impedance curves for 15 foot vehicle mounted Whip Antenna
SC-A-46691	RF Cable Assembly Tests

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction (See 3.31).- The equipment shall be constructed in accordance with the requirements of this Specification, and of Drawings DL-SC-B-685200 and DL-SC-B-685201 (including the requirements for parts, materials, and processes thereon).

3.3 Electrical installation.-

3.3.1 Power system wiring and components installation (See 4.8.2).- Power wires and components shall be installed and wired in accordance with the drawings and this Specification (See 3.31). The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- The point to point resistance of all power line conductors shall not exceed .35 ohms when tested in accordance with 4.8.2.1.

3.3.1.2 Power distributional and operational functions.- Main power branch, branch distribution and circuit switch functions thereof shall meet the operational requirements as shown by Power Schematic Drawing SC-F-595036 when tested in accordance with 4.8.2.2.

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3.3.2 Signal (audio) and Radio Frequency (rf) system wiring and components installation (See 4.8.3).- Signal (audio) and RF wires, cables and wiring components shall be installed and wired in accordance with the drawings and this Specification (See 3.3). The completed in-place signal system installation shall meet the following requirements:

3.3.2.1 Continuity of conductors.- The resistance between both end terminations of all signal and rf line conductors shall not exceed 0.5 ohms when tested in accordance with 4.8.3.1.

3.3.2.2 Insulation resistance.- The insulation resistance shall not be less than 100 megohms when tested in accordance with 4.8.3.2.

3.4 Systems operation.- With all operational equipments of the AN/GRC-122( ) assemblage installed in the shelter facility, the operational requirements of all power, audio, signal and radio frequency cable assemblies, switches, connectors, breakers and indicator units, shall be met when tested in accordance with 4.9. Specified audio frequency and radio frequency characteristics of Radio Set AN/GRC-106( ) shall be unaffected by integration of this radio set into the S-502( )/GRC-142( ) facility. Transmission power output shall be 400 watts PEP (peak envelope power). Also, integration of all other operational equipments in the shelter facility shall have no affect on their operational characteristics. During all modes of operation involving radio teletypewriter operation, not more than one error per twenty lines of "FOX" copy (See 4.8.1i) will be allowed. Transmission and reception of voice messages shall be clear and free of interfering noise or tones.

3.5 High temperature.- When tested in accordance with 4.10, the equipment shall meet the following requirements:

a. Operational check.- Transmission and reception of radio voice and radio teletypewriter test messages (See 4.8.1f & i) to a test station receiving facility at one spot frequency in the range of 2.0 to 29.9 MHz shall be checked to determine quality of voice and radio teletypewriter copy in all the operating modes listed below:

(1) 85 Hz and 850 Hz frequency shift keyed radio teletypewriter operation.

(2) Compatible AM and SSB voice operation.

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(3) Simultaneous 85 Hz frequency shift keyed radio teletypewriter and SSB voice operation.

(4) 85 Hz frequency diversity, frequency shift keyed radio teletypewriter operation.

(5) CW operation.

When received the voice message shall be clear and audible. The teletypewriter message as received on a TT-98( )/FG shall have no more than one error per twenty lines of "FOX" copy (See 4.8.1i).

b. RF power output (dummy antenna).- The rf power output shall be measured, unless otherwise specified, using the standard conditions of 4.8.1. All RF power output measurements shall be made as a function of the voltage across the resistive element of the dummy antenna (See 4.8.1c) and shall be corrected for temperature and frequency characteristics. Correction curves for frequency effects shall be furnished by the contractor for all rf meters used on these tests. RF power output shall be measured at 2.0, 5.0, 8.0, 11.5, 14.5, 16.99, 18.0, 21.5, 25.5 and 29.99 MHz, unless otherwise specified by the Government. The installed AN/GRC-106( ) Radio Set shall have the following level of power output into the 50 ohm load (dummy antenna) over the frequency range of 2.0 to 29.999 MHz for the following modes of emission:

(1) SSB Voice.- When the equipment is modulated by the standard voice test message of 4.8.1f, the equipment shall be capable of a peak envelope power output of at least 400 watts. When modulated by two equal sinusoidal audio tones (4.8.1h) the peak envelope power output shall be 400 watts (+2, -1db).

(2) Compatible AM.- The peak envelope power output of the compatible AM signal (carrier and upper side band) when modulated by the standard voice test message of 4.8.1f shall be at least 400 watts. The carrier power shall be 70 watts (+2, -1db).

(3) CW, NSK (85 Hz + voice and 85 Hz Diversity) and FSK (85 Hz and 850 Hz).- The average power output shall be 200 watts (+2, -1db).

c. Simplex operation.- Simplex (one way reversible) operation using Radio Set AN/GRC-106( ), Radio Teletype Modem MD-522( )/GRC, Teletypewriter TT-98( )/FG, and Reperforator-Transmitter TT-76( )/GGC shall provide for the "On line" transmission of teletype

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signals using the keyboard of TT-98( )/FG and/or TT-76( )/GGC with monitoring of transmitted message on the page printer of the TT-98 ( )/FG.

d. Hum and noise.- The hum and noise output of the installed AN/GRC-106( ) Radio Set when measured at the antenna terminals of the equipment shall have a value of more than 50 db below either of the two equal output frequencies corresponding to two audio input modulating tones with frequencies of 1500 and 2500 HZ when the AN/GRC-106( ) is operating at any power level from 100 watts PEP up to and including rated peak envelope power output in the SSB mode at any frequency in the range 2.0 to 29.999 MHZ. This requirement shall not supersede any requirement of MIL-STD-461A with Notice 4.

e. RF power output (dummy whip antenna).- When the equipment is operated in the SSB Voice mode and is modulated by two equal sinusoidal audio tones (4.8.1h), the peak envelop power output shall be at least 84.5 watts at 2.0 MHz.

f. Temperature extremes for the AN/GRC-106( ) exhaust and shelter internal ambient.- During all operation called for in 4.10, the temperature at the AM-3349( )/GRC-106 heat exchanger exhaust port shall not exceed 200°F and the temperature above the operators chair shall not exceed 131°F.

3.6 Humidity.- During testing in accordance with 4.11 the shelter facility shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts or materials and operation of equipment shall be unaffected by the test.

3.7 Vibration, operation.- When tested in accordance with 4.12 the listed equipments shall not suffer any functional or physical damage.

3.8 Bounce.- When tested in accordance with 4.13 the designated equipments shall not suffer any functional or physical damage. Minor paint abrasion due to this test is permissible.

3.9 Shock, bench handling.- When tested in accordance with 4.14 the designated equipments shall not suffer any functional or physical damage. Minor paint abraision due to this test is permissible.

3.10 Water testing.-

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3.10.1 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.15.1. During and after the test there shall be no evidence of leakage into the shelter facility or its walls, floor, roof or door.

3.10.2 Fording.- The shelter facility shall be capable of withstanding the test of 4.15.2. During and after the test there shall be no evidence of leakage into the shelter facility or its walls, floor, or door.

3.11 Air-Tightness.- The shelter facility shall show no evidence of air leakage through sealed seams, joints or rivets of exterior surfaces modified per DL-SC-B-685204 when tested per 4.16.

3.12 Vehicular transportation.- The shelter facility loaded and tested in accordance with 4.17 shall be capable of being transported over cross-country terrain by military vehicles without sustaining any buckling, cracking, delamination, or other permanent deformation or degradation.

3.13 Rail transportation.- The shelter facility loaded and tested in accordance with 4.18 shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail without sustaining any buckling, cracking, delamination, or other permanent deformation or degradation.

3.14 Fungus.- The equipment shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.19. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.15 Salt Fog.- After testing as specified in 4.20, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise lossened finish or metal surface.

3.16 Electromagnetic interference characteristics.- The shelter facility shall meet the electromagnetic interference requirements specified in MIL-STD-461A with Notice 4. Tests and test methods shall be as specified in 4.21.

3.17 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.22. There shall be no damage or permanent distortion to the shelter facility mounting

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surface or insert head or threads and the insert shall not have rotted from its original position. Insert threads shall be free of paint.

3.18 Cable assemblies.- (See 4.4)

3.18.1 Cable assemblies, power and audio frequency.- Radio frequency cable assemblies shall be fabricated in accordance with the drawings, unless otherwise specified, and shall meet the requirements and tests of MIL-C-3885.

3.18.2 Cable assemblies, radio frequency.- Radio frequency cable assemblies shall be fabricated in accordance with the drawings, unless otherwise specified, and shall meet the requirements and test as indicated on Drawing SC-A-46691.

3.19 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale and other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units. (See 3.31)

3.20 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 3.31 and 4.4).

3.21 Marking.- (See 3.31 and 4.4)

3.21.1 General.- Marking shall conform to Specification MIL-M-13231 and the equipment drawings.

3.21.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.21.3 Nameplates.- Nameplates shall be provided on equipments as indicated by the equipment drawings.

3.21.4 Serial Numbers.- Serial numbers shall be provided on equipments as indicated by the equipment drawings.

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3.21.5 Aircraft loading and data plates.- Aircraft loading and data plates shall be provided on the equipment as indicated by the equipment drawings.

3.22 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.31)

3.23 Seal and vapor.- All joints of the framing for the air conditioner duct ports openings, the power and signal entrance box, etc. shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be a one part sealant (silicone construction sealant SE-1204, as made by General Electric Corp. or equal). (See 3.31)

3.24 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components in the shelter securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subject to the specified transport tests. (See 3.31)

3.24.1 Screws, threaded devices and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.25 Soldering.- (See 3.31)

3.25.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.25.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come into contact with insulation material. Where acid or acid salts are used as permitted above, they shall be completely neutralized and removed immediately after use.

3.25.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting

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actions, and shall not be crystallized, over or under heated, and the minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.26 Tropicalization of material.- (See 3.31) Materials which are not fungus inert per Requirement 4 of MIL-STD-454 shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows:

3.26.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated materials will be held to the practical minimum.

3.26.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.26.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.26.4 Statement of treatment.- The contractor shall submit for approval, to the Contracting Officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2k).

3.27 Welding.- Welded joints shall be as defined by the drawing and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs,

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blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4 and 3.31)

3.28 Wiring and cabling.- Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy. (See 3.31)

3.28.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.28.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.28.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.28.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.23)

3.28.5 Identification of wiring.- Wiring shall be identified by colorcoding and wire pair marking, as indicated on the drawings.

3.29 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. Also see 4.5)

3.30 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equip-

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ment. (See 4.24) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.31 Workmanship.- The equipment shall be manufactured and assembled in accordance with applicable portions of the following paragraphs and the other detailed requirements of this specification. (See 4.25)

3.2 Construction
3.19 Cleaning
3.20 Finish
3.21 Marking
3.22 Riveting
3.23 Seal, vapor
3.24 Securing of parts
3.25 Soldering
3.26 Tropicalization
3.27 Welding
3.28 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include packaging for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment packaging for delivery. (See 4.5)

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(2) Quality conformance inspection of packaging for delivery. (See 4.26)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units.- Unless otherwise specified in the contract or purchase order the contractor shall furnish two (2) first article units of the complete Shelter, Electrical Equipment AN/GRC-122 and five specimens of each combination of wire and terminal. (See Solderless-Terminal-Lug-Connections, paragraph 3.28 and 4.23)

4.3.2 First article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data.- The first article test plan(s) and test reports(s) shall be as required in the contract or purchase order.

TABLE I - First Article Inspection

Unit One	Unit Two
Inspection by Subsidiary Documents	Inspection by Subsidiary Documents
Mounting Inserts	Mounting Inserts
Solderless Terminal Lug Connections	Solderless Terminal Lug Connections
Interchangeability	Interchangeability
Visual and Mechanical	Visual and Mechanical
Power System Wiring and Components Installation	Vibration, Operating
Signal and RF System Wiring and Components Installation	Airtightness
Systems Operation	Bounce
Airtightness	Bench Handling
EMI	Power System Wiring and Components Installation
High Temperature	Signal and RF System Wiring and Components Installation
Humidity	Systems Operation
Watertightness	Vehicular Transport
Fording	Rail Transport
Fungus *	Watertightness
Salt Fog *	Fording

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TABLE I - First Article Inspection (continued)

Unit One	Unit Two
Power System Wiring and Components Installation	Power System Wiring and Components Installation
Signal and RF System Wiring and Components Installation	Signal and RF System Wiring and Components Installation
Systems Operation	Systems Operation

* The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.4 Inspections covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this Specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Cable assemblies	3.18
Cleaning	3.19
Finish	3.20
Marking	3.21
Welding	3.27

4.5 Quality conformance inspection of equipment before packaging.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4 and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

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TABLE II - Group A Inspection

Inspection	Reqt Para	Insp Para	AQL % Major	AQL% Minor
Visual and Mechanical	3.31	4.25	1.0	4.0
Electrical				**
Power System Wiring and Components Installation	3.3.1	4.8.2		
Continuity of Conductors	3.3.1.1	4.8.2.1		1.0% for entire group
Power Distributional and Operational Functions	3.3.1.2	4.8.2.2		
Signal and Radio Frequency Systems Wiring and Components Installation	3.3.2	4.8.3		
Continuity of Conductors	3.3.2.1	4.8.3.1		
Insulation Resistance	3.3.2.2	4.8.3.2		
System Operation	3.4	4.9		

** All electrical defects are major.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection, except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III, shall be as follows:

AQL	Inspection Level
4.0%	S-4
2.5%	S-4

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4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in the order listed in Table III.

TABLE III - Group B Inspection

Inspection	Reqt Para	Insp Para	AQL
Interchangeability	3.30	4.24	4.0%
Mounting Inserts	3.17	4.22	4.0%
Air Tightness	3.11	4.16	2.5%

4.5.3 Group C inspection.- Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.-

4.5.3.1.1 Subgroup I.- This subgroup applies only to the first Group C inspection. For this subgroup, two units shall be randomly selected from the first ten (10) units produced. Selection of the units will be made by the Government. The order of testing and the distribution of tests on each unit shall be as listed in Table IV.

4.5.3.1.2 Subgroup II.- For this subgroup sampling shall be as follows:

a. Solderless Terminal Lug Connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one (1) foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

b. Vibration.- One unit shall be randomly selected from the last production lot offered for acceptance.

c. Rail Transport.- Subsequent to the first ten (10) units produced, one unit shall be randomly selected from each additional 200 units or fraction thereof produced.

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4.5.3.1.3 Subgroup III.- The equipment Verification Review, in accordance with Paragraph 4.7, shall be performed once on each part and assembly, randomly selected, and on their installation into one of the first 10 shelter facilities produced. This inspection shall be performed as the selected unit is being assembled.

4.5.3.2 Order of inspection within Group C.- The order of Group C inspection shall be in accordance with Notes 1 and 2 of Table IV.

4.5.3.3 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass the Visual and Mechanical inspection of 4.25.

TABLE IV - Group C Inspection

Inspection	Reqt Para	Insp Para
Subgroup I (Initial Group C only)		
Unit One		
High Temperature	3.5	4.10
Humidity (See Note 3)	3.6	4.11
Fungus (See Note 6)	3.14	4.19
Salt Fog (See Notes 4 & 6)	3.15	4.20
Watertightness	3.10.1	4.15.1
Fording	3.10.2	4.15.2
Unit Two		
Vibration	3.7	4.12
Bounce	3.8	4.13
Shock, Bench Handling	3.9	4.14
Systems Operation	3.4	4.9
Vehicular Transport	3.12	4.17
Rail Transport (See Note 5)	3.13	4.18

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TABLE IV - Group C Inspection (continued)

Inspection	Reqt Para	Insp Para
Watertightness	3.10.1	4.15.1
Fording	3.10.2	4.15.2
Subgroup II (Subsequent Group C Inspections)		
Solderless Terminal Lug Connections	3.28.4	4.23
Vibration, Operation	3.7	4.12
Systems Operation	3.4	4.9
Rail Transport (See Note 5)	3.13	4.18
Subgroup III		
Equipment Verification Review	3.2	4.7

NOTES:

1. The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.5.3.

2. The order of inspection within subgroups of Group C shall be as listed in Table IV.

3. After subjection to the Humidity test the unit shall be subjected to and meet the requirements of the watertightness and fording tests of 4.15 and 4.16 prior to performing any additional testing.

4. After subjection to the Salt Fog test the unit shall be subjected to and meet the requirements of the Power System Wiring and Components Installation, Signal and RF System Wiring and Components Installation, and Systems Operation tests of 4.8.2, 4.8.3 and 4.9 respectively.

5. After subjection to the Rail Transport test the unit shall be subjected to and meet the requirements of the Watertightness, Fording, Power System Wiring and Components Installation, Signal and RF System Wiring and Components Installation and Systems Operation tests of 4.8.2, 4.8.3, 4.9, 4.15.1 and 4.15.2.

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6. The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.6 Bounce preconditioning.- The unit, with shock mounts (if any) removed or blocked shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.29)

4.7 Equipment verification review.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on contract or order against the drawings cited for construction (para 3.2). The EVR shall consist of the following:

a. An audit to establish that "as-built" configuration of units, assemlbies, modules, or parts) are in accordance with the end product drawing requirements.

b. An audit to establish that "as-built" configuration of units, assemblies, modules, or parts meet the acceptance requirements specified in the technical data for each unit, assembly, module, or part. The audit shall record all acceptance test methods and inspection equipment used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of unit, assembly, module, or part and details proving compliance with 4.7a and 4.7b.

(2) Discrepancies noted.

(3) Corrective action taken.

d. Four (4) copies of the EVR report shall be submitted to the Government Technical organization cited elsewhere in the contract for submission of ECP's and one (1) copy shall be furnished to the DCAS inspector, not later than 15 days prior to the submission for inspection of the first production unit or lot. Government evaluation of

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the results of EVR will be to the depth and extent necessary to demonstrate that the "as-built" hardware and quality assurance provisions are in accordance with the cited drawing requirements and specifications. In the event Government evaluation reveals discrepancies between the "as-built" hardware and the cited drawings, all equipment acceptance shall stop until such time as all discrepancies have been resolved or corrected. The contractor shall provide the following:

(1) Access to test stations and facilities employed by the contractor during in-process and final inspection.

(2) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(3) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(4) Adequate administrative support for the EVR.

4.8 Electrical installation tests.- The tests of 4.8.2 through 4.8.3.2 shall be performed on each shelter facility. Before applying AC or DC power to the shelter facility, an electrical ground at the power entrance box shall be made.

4.8.1 Standard test conditions.- Unless otherwise specified, the following standard test conditions shall apply during all testing:

a. Atmosphere.- All tests shall be made at prevailing factory conditions of temperature, humidity, and pressure, unless otherwise specified in the detailed procedure.

b. 115 VAC and 28 VDC.- Both DC and AC primary power sources are required for operation during testing. (See 4.8.1j and k)

c. Dummy antenna.- A 50-ohm resistive load shall be used except where the detailed procuedure requires the use of a whip antenna. The 50-ohm antenna shall be non-reactive, and shall have a continuous duty rating of 500 watts.

d. Whip Antenna.- Where the detailed testing procedure requires a whip antenna, a dummy ship antenna shall be used unless otherwise specified herein. The dummy antenna characteristics must conform to Drawing SC-C-4500 and the characteristics as measured

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shall be submitted to the contracting officer's technical representative for approval at least 90 days prior to any testing.

e. Grounding.- A single ground connection shall be provided which shall connect to the internal grounding system of the equipment. Test equipment shall be grounded to the common equipment ground wherever practicable. The cases of metal cased test equipment shall be grounded and so noted in the test record.

f. Standard voice test message.- The contractor will furnish a magnetic tape upon which is recorded a test message to be used where specified in the detailed test procedure. The test message will consist of the following phrase repeated over and over: "Jo took father's shoe bench out, she was waiting at my lawn". The test message signal level shall be an average 200 MV (RMS)/600 ohms.

g. Warm-up period.- Unless otherwise specified, measurements shall be made, after five minutes from the time primary power is applied.

h. Standard two-tone input.- Unless otherwise specified in the detailed test procedure, the modulation applied to the equipment shall be two audio tones equal in amplitude (200 MV (RMS)/600 ohms) and whose frequencies shall be 1500 and 2500 Hz.

i. Standard teletype (FOX) message.- The contractor shall furnish a paper tape upon which is recorded a test message to be used where required in the detailed test procedure. The test message shall consist of the following phrase repeated over and over: "The quick brown fox jumped over the lazy dog's back; 1, 2, 3, 4, 5, 6, 7, 8, 9, 0."

j. DC primary power.- The power source shall be 28.0 ± 0.3 volts, direct current (VDC) with a 100 ampere capacity except where specifically changed in the detailed procedure. Noise and ripple voltage on the source shall be less than 0.3 volts, AC. The regulation shall be such as to maintain this voltage when the load is changed from no load to the maximum for any of the equipments under test. The voltage source shall be adjustable over the range of 24 to 30 volts DC.

k. AC primary power.- The power source shall be 115 ± 2 volts alternating current (VAC), single phase, 47 to 63 Hz, with a 60 ampere capacity except where specifically changed in the detailed procedure. The regulation shall be such as to maintain this voltage when the load is changed from no load to the maximum for any of the equipments under test.

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4.8.2 Power system wiring and components installation tests.- Perform the tests as follows:

4.8.2.1 Continuity of conductors test.- Continuity resistance tests to determine conformance with the requirements of 3.3.1.1 shall be performed on each power circuit in accordance with the paragraphs below. A calibrated low resistance reading ohmmeter or equivalent shall be used in performing the test. (Refer to Drawing SC-F-595036).

a. Place the following circuits in the ON position.

(1) Main circuit breaker CB4

(2) Power supply circuit breaker CB5

(3) DC main circuit breaker CB1

(4) OWR circuit breaker CB2

(5) DX circuit breaker CB3

(6) 28 VDC CONV circuit breaker CB6

(7) Blowers and lights circuit breaker CB7

b. Place AC/DC control switch to AC position.

c. Measure continuity resistance from pin A of AC power input connector located at power entrance box to the following:

(1) Pin X of connector J5 at Power Terminal Assembly.

(2) Pin B of connectors J1 thru J4 at Power Terminal Assembly.

(3) Pin C of 115 VAC AUX Equipment Connectors attached to curbside and roadside rack holding auxiliary equipment.

(4) Brass contacts of 115 VAC CONV outlets located at curbside raceway.

d. Place Power Supply and Main Circuit Breaker CB5 and CB4 in the OFF position.

e. Measure continuity resistance from Pin B of AC power input connector to the following:

(1) Pin Y of connector J5 at Power Terminal Assembly.

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(2) Pins S of connector J1 thru J4 at Power Terminal Assembly Box.

(3) Pin A of 115 VAC AUX equipment connectors attached to rack holding auxiliary equipment.

(4) Silver contacts of 115 VAC CONV outlets located at curbside.

f. Measure continuity resistance from Pin C of AC power input connector to the ground stud (E31) at the Power Entrance Box.

g. Place AC/DC control switch to DC position.

h. Measure continuity resistance from Pin A of DC power input connector to the following:

(1) Pin A of PWR IN connector to heater.

(2) Brass contact of 28 VDC CONV outlet located at curbside raceway.

(3) Pin A of 28 VDC AUX equipment connectors attached to interconnecting boxes for auxiliary equipment.

(4) Pins A and B of Motor Generator connectors J6 and J7 located at Power Terminal Assembly Box.

(5) Pins A and B of power connector to Amplifier AM-3349( )/GRC-106, Receiver-Transmitter RT-834( )/GRC-106, Auxiliary Receiver Transmitter RT-834( )/GRC-106, and Radio Teletypewriter Modem MD-522( )/GRC.

i. Place all remaining circuit breakers in the OFF position.

j. Measure continuity resistance from Pin C of DC power input connector to the following:

(1) Pin D of DC power input connector.

(2) Ground stud (E31) at Power Entrance Box.

(3) Silver contact of 28 VDC CONV outlet.

(4) Pin C of PWR-IN connector to heater.

(5) Pin W of connector J5 at Power Terminal Assembly Box.

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(6) Pin G of connectors J1 thru J4 at Power Terminal Assembly Box.

(7) Pins C and D of connectors J6 and J7 at Power Terminal Assembly Box.

(8) Pins C and D of power connectors to AM-3349( )/GRC-106, RT-834( )/GRC-106, AUX RT-834( )/GRC-106 and MD-522( )/GRC.

(9) Pin C of 28 VDC AUX equipment connectors attached to interconnecting boxes for auxiliary equipment.

4.8.2.2 Power distributional and operational functions test.- Power distributional and operational functions tests to determine conformance with the requirements of 3.3.1.2 shall be performed on each power circuit in accordance with the following paragraphs: (Refer to Drawing SC-F-595036)

4.8.2.2.1 Test preparation.-

a. Place 115 volt AC test lamps with grounding plug into each 115 VAC conveniences and equipment outlets.

b. Place 28 volt DC test lamps into the 28 VDC convenience outlet and equipment outlets.

c. Place all circuit breakers in the OFF position.

d. Place AC/DC control switch to DC position.

e. Place blower switch S1 to the OFF position.

f. Place ventilation switch for AN/GRC-106( ) to the OFF position.

g. Open shelter facility door.

4.8.2.2.2 Test of AC Power line circuits.-

a. Apply 115 volts AC (nominal), 60 Hertz, single phase power to the shelter facility at AC Power IN connector. Voltmeter inside shelter facility shall indicate 115 VAC $\pm$ 2%.

b. Place MAIN circuit breaker CB4 in the ON position. AC power indicator lamp shall light.

c. Test lamps in 115V Conv Rec shall light.

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d. Place AC/DC control switch to AC position. AC test lamp on connectors J1 thru J4 of Power Terminal Assembly shall light.

e. Measure AUX equipment AC line voltage by connecting a calibrated AC voltmeter (0-150 volt scale) across Pins A and C of each 115 VAC AUX equipment power connector. Voltage measured shall be 115 VAC $\pm$ 2%.

f. Place PWR SUP circuit breaker CB5 in the ON position. The test lamp on connector J5 of Power Terminal Assembly box shall light.

g. Place Blowers and Lights circuit breaker CB7 in the ON position.

h. Close shelter facility door. DC incandescent lamps shall light. Open shelter facility door, incandescent lamps shall go out.

i. Place blower switch S1 in the ON position. Front wall right hand blower shall operate and exhaust air from shelter facility.

j. Close shelter facility door to restore incandescent lamps.

k. Place Blowers and Lights circuit breaker (CB7) in the OFF position. Exhaust blower shall cease to operate. Incandescent lamps shall go out.

l. Open shelter facility door.

m. Place PWR SUP circuit breaker CB5 in the OFF position. Remove AC test lamp at connector J5 of Power Terminal Assembly and replace with AC line plug of Power Supply PP-4763( )/GRC.

n. Place PWR SUP circuit breaker CB5 in the ON position and turn on PP-4763( )/GRC.

o. Measure operating equipment DC line voltage by connecting a calibrated DC voltmeter (0-50 volt scale) across Pins A and C of each power connector to the AM-3349( )/GRC-106, RT-834( )/GRC-105, MD-522( )/GRC, and AUX RT-834( )/GRC-106. Voltage measured shall be 28 $\pm$ 2 volts DC.

p. Place ventilating switch for the AN/GRC-106( ) assist blower in the ON position. Blower shall operate and force air through the AN/GRC-106( ) Radio.

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q. Place PWR SUP (CB5) and MAIN (CB4) circuit breakers in the OFF position. All lamps shall go out. Exhaust blower for the radio shall cease to operate.

4.8.2.2.3 Test of DC Power line circuits.-

a. Apply 28 ± 2 volts DC to the shelter facility at DC power IN connector.

b. Place DC Main circuit breaker CB1 in the ON position. DC power indicator lamp shall light to indicate DC power to shelter facility.

c. Place COVN outlet circuit breaker CB6 to ON position. Test lamp in 28 VDC convenience outlet shall light.

d. Measure Heater DC line voltage by connecting a calibrated DC voltmeter (0-50) volt scale) across Pins A and C of power connector to heater. Voltage measured shall be 28 ± 2 volts DC.

e. Place AC/DC control switch to the DC position. Voltmeter shall indicate 28 ± 2 volts DC.

f. Measure Equipment and Aux Equipment DC line voltage by connecting calibrated DC voltmeter to Pins A and C of each power connector to AUX Equipment, AM-3349( )/GRC-106, RT-834( )/GRC-106, MD-522( )/GRC and AUX RT-834( )/GRC-106. Voltage measured shall be 28 ± 2 volts DC.

g. Place Blowers and Lights circuit breaker CB7 in the ON position.

h. Repeat steps 4.8.2.2.2h thru l.

i. Place ventilating switch for the radio assist blower to the OFF position. Blower shall cease to operate.

j. Place INVERTERS OWR circuit breaker CB2 in the ON position. Motor generator shall operate. AC test lamp on connectors J1 and J2 at Power Terminal Assembly shall light.

k. Place INVERTERS DX circuit breaker CB3 in the ON position. Auxiliary motor generator shall operate. AC test lamp on connectors J1 and J2 at Power Terminal Assembly shall light.

l. Place 28 VDC CONV circuit breaker CB6 in the OFF position. DC test lamp in 28 VDC CONV outlet shall go out.

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m. Place 28 VDC CONV circuit breaker CB6 to the ON position. DC test lamp in 28 VDC CONV outlet shall light.

n. Place DC main circuit breaker CB1 in the OFF position. Motor generators shall cease to operate. All lamps shall go out.

o. Place all circuit breakers in the OFF position.

4.8.3 Signal (audio) and Radio Frequency system wiring and components installation test.- Perform the tests as follows:

4.8.3.1 Continuity of conductors test.- Continuity resistance tests to determine conformance with the requirements of 3.3.2.1 shall be performed on each signal and rf circuit in accordance with the paragraphs below. A calibrated low resistance reading ohmmeter or equivalent shall be used. (Refer to Drawing SC-F-595035)

4.8.3.1.1 Test of Switch Assembly SA-1650( )/GRC-142.-

a. Perform the following circuit functions:

(1) DX-RCV PONY switch S1 to black position.

(2) OWR DX-SEND switch S2 to BLACK position.

(3) OWR DX-SEND/DX-REC PONY switch S3 to OWR DX-SEND position.

(4) LOCAL/REMOTE switch S4 to LOCAL position.

(5) LOCKOUT/OVERRIDE switch S5 to LOCKOUT position.

b. Circuit continuity resistance which meets the requirements of 3.3.2.1 shall exist between chassis ground and the following: Pin A of connectors J1 thru J10; Pins B and H of connectors J18 and J19; and Pin F of connector J16.

c. Circuit continuity resistance which meets the requirements of 3.3.2.1 shall exist when tested in accordance with Table V.

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TABLE V - Continuity Measurements

FROM CONNECTOR	PIN	TO CONNECTOR	PIN	
J1	B	J4	B	
J1	C	J2	B	
J2	C	J4	C	
J3	B	J5	B	
J6	C	J5	C	
J6	B	J9	B	68 ± 5% ohms
J6	C	J9	C	
J8	B	J7	C	
J8	C	J10	C	
J7	B	J10	B	
J11	A	J12, J13	A	
J11	C	J16	A	
J11	E	J12, J13	E	
J11	F	J12, J13	F	
J12	C	J13	C	
J12	C	J16	B	
J14	A	J14	C	
J15	A	J15	C	
J16	C	E2		
J16	E	J17	B	
J17	A	E1		
(INSERT OPENED PLUGS INTO J18 & J19)				
J18	TIP	J14	C	
J18	RING	J14	A	
J19	TIP	J15	C	
J19	RING	J15	A	
(PLACE ALL SWITCHES OF (a) ABOVE TO ALTERNATIVE POSITION AND REMOVE OPEN PLUGS FROM J18 and J.9).				
J1	B	J4	B	27 ± 5% Kohms
J3	B	J5	B	27 ± 5% Kohms
J6	B	J8	C	
J7	C	J10	C	68 ± 5% ohms
J8	B	J9	B	
J11	C	J12, J13	C	
J14	A	J17	E	

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TABLE V - Continuity Measurements (continued)

FROM CONNECTOR	PIN	TO CONNECTOR	PIN
J14	C	J17	F
J15	A	J17	C
J15	C	J17	D
J16	C	J16	E
J16	C	J17	B

4.8.3.1.2 Test of Signal Entrance Box.- Circuit continuity resistance which meets the requirements of 3.3.2.1 shall exist when tested in accordance with Table VI.

TABLE VI - Continuity Measurements

FROM BINDING POST	TO CABLE	PIN/WIRE
E3	W-26	WHT
E4	W-26	WHT/BRN
E1	CX-1852( )/U	H
E2	CX-1852( )/U	F
E5	W-15	F
E6	W-15	E
E7	W-15	D
E8	W-15	C
E9	W-15	B
E10	W-15	A

4.8.3.1.3 Test of Remaining Signal Lines.- Circuit continuity resistance which meets the requirements of 3.3.2.1 shall exist when tested in accordance with the following paragraphs:

a. For cables W1 thru W3: Pin A to shield, Pin B to white wire, and Pin C to black wire.

b. For cables W4 thru W6: Pin A to shield, Pin B to black wire, and Pin C to white wire.

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c. For cables W7 thru W10 and W14: Pin A to Pin A, Pin B to Pin B, and Pin C to Pin C.

d. For cables W11 and W12: Sleeve to black plug, tip to blue plug, and ring to white plug.

e. For cable W15: Pin A to white wire, Pin B to white/brown wire, Pin C to white/red wire, Pin D to white/green wire, Pin E to white/orange wire, and Pin F to white/black wire.

f. For cables W17 and W18: Pin A to Pin A, Pin B to Pin B, Pin C to Pin C, Pin D to Pin D, Pin E to Pin E, Pin F to Pin F, Pin J to Pin J (W17 only), and Pin H to Pin H.

4.8.3.2 Insulation resistance tests.- Insulation resistance tests to determine conformance with the requirements of 3.3.2.2 shall be performed on each signal and RF line circuit in accordance with the paragraphs below. A non-destructive insulation resistance tester or equivalent shall be used in performing the test. (Refer to Drawing SC-F-595035)

NOTE: Take preventive measures that will preclude personnel from coming into contact with conductors under test during performance of ALL insulation resistance tests. Discharge each conductor to ground following each test, to eliminate danger of shock.

4.8.3.2.1 Test of Switch Assembly SA-1650( )/GRC-142.-

a. Ground switch assembly SA-1650( )/GRC-142 to proper ground terminal.

b. Remove CALL lamp from switch assembly.

c. Connect all pins for connector J1 thru J19 to a common ground terminal using appropriate test jigs or test fixtures.

d. Perform the circuit functions listed under 4.8.3.1.1a.

e. Connect tester between ground terminal and Pin B of J1. Remove ground connection from Pin B of J4.

f. Adjust tester to a value of not less than 100 volts DC output and maintain this potential until the meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.2.2 shall be met. Re-establish ground connections to the pins under test.

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g. Repeat steps e and f above for each pin listed under Table VII, each time ungrounding the pins correlating with the circuit being tested and reconnecting to ground following completion of tests.

TABLE VII - Continuity Measurements

CONNECTOR	PIN TESTED	CONNECTOR	PIN UNGROUNDED
J1	C	J2	B
J2	C	J4	C
J3	B	J5	B
J3	C	J5	C
J6	B	J9	B
J6	C	J9	C
J7	B	J10	B
J7	C	J8	B
J8	C	J10	C
J11	A	J12, J13	A
J11	C	J12, J13, J16, J16	C, C, A, B
J11	E	J12, J13	E
J11	F	J12, J13	F
J16	C	E2	-
J16	D	-	-
J16	E	J17	B
J17	A	E1	-
J17	C	-	-
J17	D	-	-
J17	E	-	-
J17	F	-	-
J14	A	J14	C
J15	A	J15	C

4.8.3.2.2 Test of Remaining Signal Lines.-

a. Position or isolate all disconnected cable assemblies so that the conductors do not short to each other or to shelter facility ground.

b. Connect all pins of cable W1 to a common ground terminal.

c. Connect tester bewteen Pin A and the ground terminal. Remove ground connection from Pin A. Adjust tester to a value of not less than 100 volts DC output and maintain this potential until the meter

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has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.2.2 shall be met. Re-establish ground connection to the pin under test.

d. Repeat steps b and c above for each of the remaining pins or conductors of the same cable; each time ungrounding the pin correlating with the circuit being tested and reconnecting to ground following completion of test.

e. Repeat steps b thru d above for each remaining cable assemblies W2 thru W14, W17 and W18.

f. Connect tester between binding post E1 and shelter facility ground terminal. Adjust tester to a value of not less than 100 volts DC output and maintain this potential until the meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirements of 3.3.2.2 shall be met.

g. Repeat step f above for each remaining binding post E2 thru E10.

4.9 Systems operation test.- Install all operational equipments comprising the AN/GRC-122( ) in the shelter facility and make all electrical and mechanical connections. Perform the tests below to demonstrate compliance with the requirements of 3.4.

4.9.1 Half duplex (one way reversible) operational test.- Perform the test as follows:

a. Set up three stations which shall be referred to as the remote site, the test station, and the assemblage (This is the shelter facility equipped as an AN/GRC-122( )) being tested. The remote site shall be equipped with Teletypewriter TT-98( )/FG, (set for 20 MA), Telephone Set TA-312( )/PT, Telegraph Key KY-116( )/U, Control Group AN/GRA-6( ), and Remote Box C-7279( )/GRC-142. Connect the remote site to the assemblage being tested using WD-1 field wire per the step by step procedure given in the AN/GRC-142( ) Operators Guide (This information can also be obtained from the contracting officers technical representative). Set the Control Panel Local-Remote switch to Remote for all remote teletypewriter operation. Set up the test station as another functional AN/GRC-122( ) and position it 25 feet away from the assemblage being tested. Use dummy whip antenna per 4.8.1c on the test station, unless otherwise specified. Operation of the assemblage shall be from both AC and DC primary power. The particular primary power to be used shall be designated below by the required position of the AN/DC control switch on the Power Distribution Panel SB-3358( )/GRC-142. It shall be necessary to operate the test station only from AC primary power.

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b. Set test station AN/GRC-106( ) and AN/GRC-106( ) in assemblage tested to a frequency in the range 2.0 to 29.999MHz.

c. Perform starting procedures per operator's guide. Make initial checks using AC primary power.

d. Perform tuning procedures per operator's guide, observing general operating notes of operator's guide.

e. Set controls at test station, and at assemblage being tested, for VOICE operation per operator's guide.

NOTE: During each transmission at an assigned frequency, make certain that the frequency is clear before transmitting.

f. Transmit a long count to, and then receive a long count from the test station. The transmission and reception shall be clear and free of interfering noise or tones. After making test, transmit and receive a long count at remote site, using AN/GRA-6( ).

g. Place AC/DC control switch to DC position and repeat step f above.

h. Place AC/DC control switch to AC.

i. Set controls at test station and at assemblage being tested, for 850 Hz operation per operator's guide.

j. Transmit twenty lines of "FOX" copy to an then from test station. The requirements of 3.4 shall be met. Transmission shall be on the TT-98( )/FG; reception on the TT-76( )/GGC. Check TT-76( )/GGC tape reader, tape punch, and keyboard and TT-98( )/FG for proper operation.

k. Switch LOCAL/REMOTE switch to REMOTE and repeat step j above at the remote site.

l. Place AC/DC switch to DC and repeat steps j and k above for local and remote RTTY operation using 850 Hz mode.

m. Place AC/DC switch to AC.

n. Set controls at test station and at assemblage being tested, for 85 Hz operation per operator's guide.

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o. Repeat tests of steps j thru m above for 85 Hz operation.

p. Set controls at test station and at assemblage being tested, for 85 Hz + VOICE operation per operator's guide.

q. Repeat tests of steps f and g, j thru m above for 85 Hz + VOICE operation. Check for teletype (TTY) errors when microphone switch is operated.

r. Set controls at test station and at assemblage being tested for 85 Hz DIV operation per operator's guide.

s. Repeat tests of steps j thru m above for 85 Hz DIV operation. Check for proper scope indication in 85 Hz DIV.

t. Disconnect whip antenna lead in. Connect coaxial cable from Meter ME-165( )/G to the 50 OHM LINE receptacle on the AN/GRC-106( ). Tune AN/GRC-106( ) into ME-165( )/G dummy load and measure CW power output after tuning procedure is completed. (See operator's guide). CW power output should be 200 watts nominal. (159 watts minimum; 317 watts maximum).

u. Measure power output at four other widely separated frequencies, including 2.1 MHz, 14.3 MHz and 28.5 MHz per procedure of step t above.

4.9.2 Full duplex operational test.- Perform the test as follows:

a. Set up the three stations in accordance with 4.9.1a except that a duplex teletypewriter TT-98( )/FG shall also be provided at the remote site. Connect the DC signal line to the assemblage at the DX RCV/PONY binding posts using WD-1 field wire.

b. Switch Modem MD-522( )/GRC ONE-WAY-DUPLEX switch to DUPLEX. Connect auxiliary LS-166( )/U to unused AUDIO receptacle on auxiliary RT-834( )/GRC-106.

c. Check operation of auxiliary RT-834( )/GRC-106 using short piece of hookup wire as an antenna connected to exterior half of bulkhead receptacle on door end of the assemblage. Use this piece of hookup wire as a receiving antenna during all full duplex tests. Use doublet (50 ohm dummy antenna) as the transmitting antenna.

d. Place AC/DC switch to DC position.

e. Check operation of auxiliary Inverter PU-724( )/U (DC operation) and auxiliary TT-98( )/FG by copying radio teletype (RTTY)

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transmission from the test station.

f. Place AC/DC switch to AC position.

g. Set AN/GRC-106( ) in assemblage under test and auxiliary receiver in test station to an assigned frequency. Set AN/GRC-106( ) in test station and auxiliary receiver in assemblage under test to another assigned frequency, 10 to 12% above or below the test stations receiving frequency.

h. Perform tuning procedures per operator's guide, observing general operating notes of operator's guide.

i. Perform tests of 4.9.1e and f for VOICE operation locally in assemblage. Adjust speaker volumn to minimize feedback in full duplex mode. The transmission and reception shall be clear and free of interfering noise or tones. DO NOT attempt full duplex VOICE operation from remote site.

j. Perform 850 Hz RTTY test of 4.9.1i thru k.

k. Replace the TA-312( )/PT at the remote site by the KY-116( )/U.

l. Disconnect the field wire from the "TA-312( )/PT" binding posts at the shelter facility and connect to the "REM CW" binding posts. Connect the CW plug inside the shelter facility to the VOICE KEY receptacle on the Switch Box SA-1650( )/GRC-142.

m. Transmit a series of "V's" spaced by one or two seconds apart from the remote site. Side tone should be heard in the AN/GRA-6( ) handset. Test station shall receive clean and accurate "V" signals in CW.

n. Turn off power per operator's guide and secure shelter facility.

4.10 High temperature.- The equipment shall be tested in accordance with Method 501, Procedure I, of MIL-STD-810 to determine compliance with the requirements of 3.5. The temperature for Step 2 shall be 155°F (+5, -0)°F. In Step 3 maintain the temperature for a minimum of 24 hours or to thermal equilibrium, whichever occurs first. The operating temperature for Step 4 shall be 120°F (+5, -0)°F. In addition to the requirements of MIL-STD-810 the conditions below shall be met:

a. The temperature in the AM-3349( )/GRC-106 heat exchanger exhaust port, between the rear of the AM-3349( )/GRC-106 and the cab end wall, one foot from the ceiling above the operator's chair, between

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the rear of the PP-4763( )/GRC and the cab end wall, and in the chamber outside the shelter shall be monitored throught the entire test.

b. Unless otherwise specified, the standard test conditions of Paragraph 4.8.1 shall apply.

c. For the purposes of this test the shelter shall be configured and operated as an AN/GRC-122( ).

d. The shelter shall be operated from AC primary power.

e. For Step 1 the equipment shall meet full specification performance for the requirements specified in 3.5a thru e.

f. For the temperature test the door of the shelter shall be closed, the door intake louvers open full, the air conditioner ports closed, the ports for the ventilating equipment open, and the intake and exhaust ports for the multi-fuel heater closed.

g. A bench setup shall be provided outside the chamber to receive transmission from the shelter being tested.

h. For Step 5 operation shall be as follows:

(1) Turn on all operational GFE equipment (except the PU-724( )/U Inverters, the multi-fuel heater, and the electric heater), the shelter lights, and the ventilating equipment for the radio and the shelter.

(2) Next, demonstrate compliance with 3.5a, b and e at 2.0 MHz.

(3) Next, set the AN/GRC-122( ) to transmit continuously on 85 Hz NSK with tape copy being transmitted through the TT-76( )/GGC with home page copy being printed on the TT-98( )/FG for a period of four (4) hours at 2.0 MHz using the dummy whip antenna. Reception shall also be received on a TT-98( )/FG at a test station. At the end of the four (4) hour key-down test demonstrate compliance with 3.5a, b and e at 2.0 MHz.

i. For Step 7 the equipment shall meet full specification performance for the requirements specified in 3.5 a thru 3.

j. During all operation the temperature at AM-3349( )/GRC-106 heat exchanger exhaust port and above the operator's chair shall meet the requirements of 3.5f.

4.11 Humidity test.- The shelter facility configured as an AN/GRC-122( ) shall be subjected to the humidity test specified in Method 507, Procedure II, of MIL-STD-810 to determine conformance to the re-

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uirements of 3.6. The chamber and accessories shall meet the requirements of Paragraph 2.1 of Method 507 except that the pH of the water shall not be less than 6.5 or greater than 7.5 at 25°C. Operation where required shall be in accordance with 4.11.1. Operation for Step 6 in Method 507, Procedure II shall be during the last five hours of each cycle.

4.11.1 Operational check during humidity.-

a. Energize the shelter using AC primary power. Operate the following equipments for a period of ten (10) minutes:

(1) AN/GRC-106( )
(2) MD-522( )/GRC
(3) AUX RT-834( )/GRC-106
(4) TT-98( )/FG
(5) AUX TT-98( )/FG
(6) TT-76( )/GGC
(7) PP-4763( )/GRC
(8) Ventilating equipment
(9) Lights
(10) Using test lamps in place of actual equipments assure proper operation of the security equipment, multi-fuel heater, and convenience outlets.

b. Next, energize the shelter using DC primary power. Operate the equipments in paragraph a above except substitute the PU-724( )/U and AUX PU-724( )/U for the PP-4763( )/GRC.

4.12 Vibration operation test.- The equipments listed in 4.12.1 shall be tested as follows to determine conformance to 3.7:

a. Secure the equipment in a test fixture, which simulates actual mounting method. This fixture shall be secured to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connections (such as cable assemblies) in place.

b. Vibrate the complete equipment assembly successively in three mutually perpendicular directions over a frequency range of 10 to 55 cps. The total excursion of the applied vibration shall be 0.040 inch in the horizontal plane and 0.060 inch in the vertical plane.

c. In each of the tree directions, change the frequency in steps of one cps.

d. The equipment shall be subjected to 30 minutes of vibration in each of three mutually perpendicular planes parallel to the equipment

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surfaces. Electrical tests in accordance with 4.12.2 shall be made before the test, while being vibrated, and after the tests.

4.12.1 Equipments to be tested.- The following equipments shall be tested:

a. Switch Box Assembly SA-1650( )/GRC-142

b. Power Distribution Panel SB-3358( )/GRC-142

c. Interconnecting Boxes J-2728( )/GRC-142

d. Remote Box C-7279( )/GRC-142

4.12.2 Electrical check during vibration.- The equipments shall be checked for electrical continuity. All power circuits shall be monitored with test lamps. The signal circuits can be checked with an ohmmeter. Continuous monitoring of all electrical circuits is required.

4.13 Bounce test.- The equipments listed in 4.12.1 shall be tested in accordance with Method 514.1, Procedure IX, Part 2 of MIL-STD-810 to determine conformance with the requirements of 3.8. In addition to the requirements as stated in Step 3 in no case shall the speed of the package tester exceed 285 revolutions per minute. For Steps 1 and 4 operation shall be in accordance with 4.12.2. In addition to the requirements of Method 514.1, Procedure IX, Part 2 the test of 4.12.2 applies during as well as before and after the bounce test.

4.14 Shock, Bench handling test.- The equipments listed in 4.12.1 shall be tested in accordance with Method 516.1, Procedure V, of MIL-STD-810 to determine conformance with the requirements of 3.9. At the conclusion of the test operation shall be in accordance with 4.12.2. Operation is not required during the test.

4.15 Water testing.-

4.15.1 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc. shall be subjected to the following two part test: (NOTE: Perform tests with inside cover of the entrance box removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, Model

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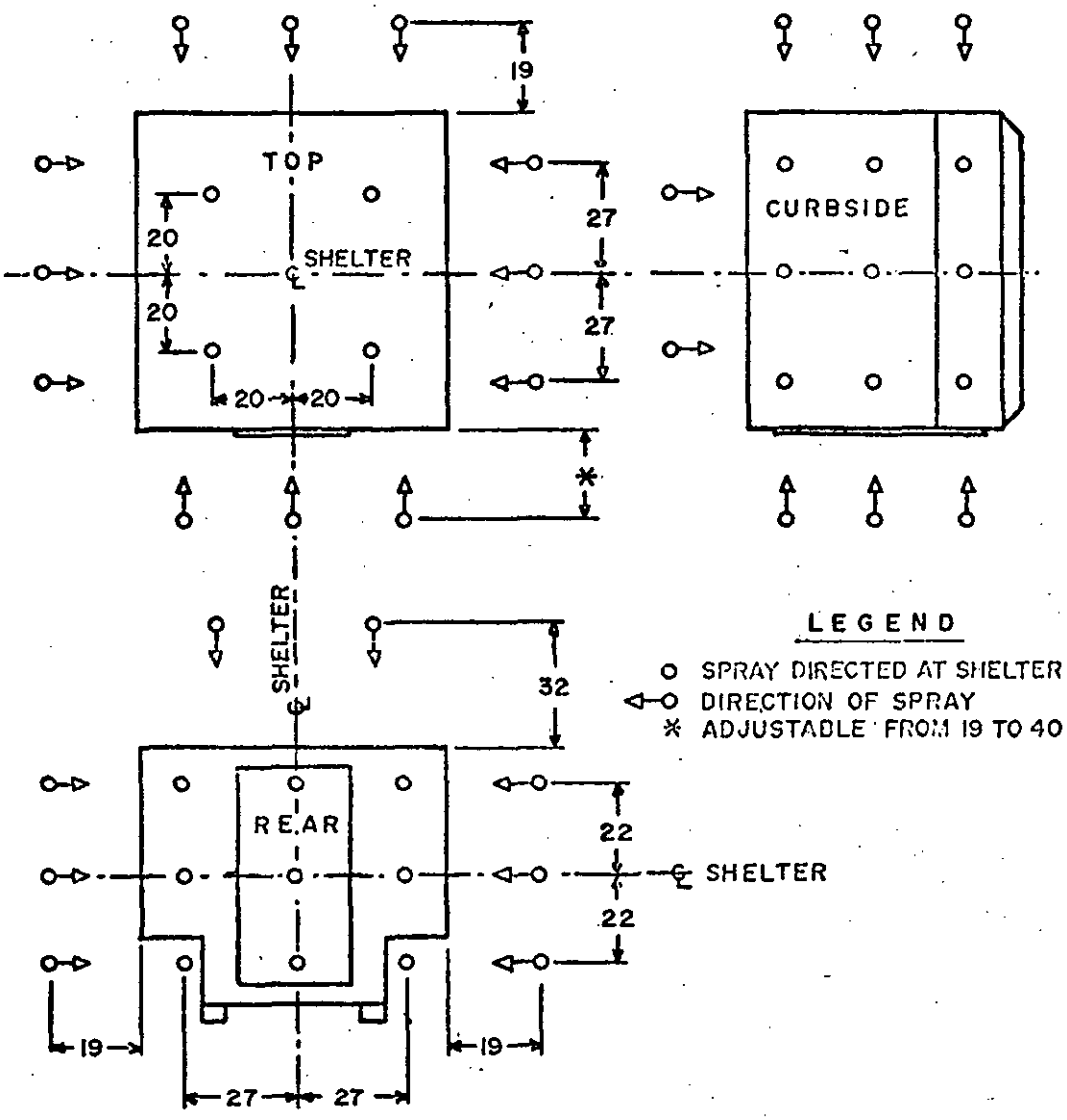
G29SQ (GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surfaces and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using nine nozzles, and the roof using four nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See Figure 1 for Nozzle location). Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the Fording test of 4.16 the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door end panel). Upon completion of the above test, the shelter facility shall meet the requirements of 3.10. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position (with exception of the shelter door vent panel and the air conditioning duct port covers of the curbside wall), and repeat paragraph a above.

4.15.2 Fording test.- The shelter facility shall be immersed in water for 1 hour to a depth of 26 inches (measured from the bottom of the skids) and meet the requirements of 3.11. No special fording kits shall be used. The shelter shall be restrained from floating by external tie downs. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the Government at the time of testing. After this determination the holes shall be sealed with rivets and sealer.

4.16 Air-Tightness test.- All exterior modifications required to configure an S-250/G shelter into an S-502( )/GRC-142( ) shall be tested to assure that air will not pass through the sealed areas. The test shall be performed by inserting air through gas heater exhaust opening by means of an appropriate fixture. The test apparatus shall include an air supply sufficient to maintain the required pressure, a means to regulate air flow and input pressure gage. A manometer graduated to a .1 inch or less, shall be utilized to measure a required 6 inches minimum of water differential pressure between the interior of the shelter and outside ambient pressure. All areas of the shelter that were modified as a result of the contract requirements will be painted with Liquid Soap and Water Solution. There shall be no leaks as evidenced by bubbling or spurting.

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NOTES

1. NOZZLES SHALL BE PERPENDICULAR TO SHELTER SURFACE.
2. UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES
3. GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES
4. ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE
5. FRONT NOZZLE LOCATIONS SAME AS REAR.
6. DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 1

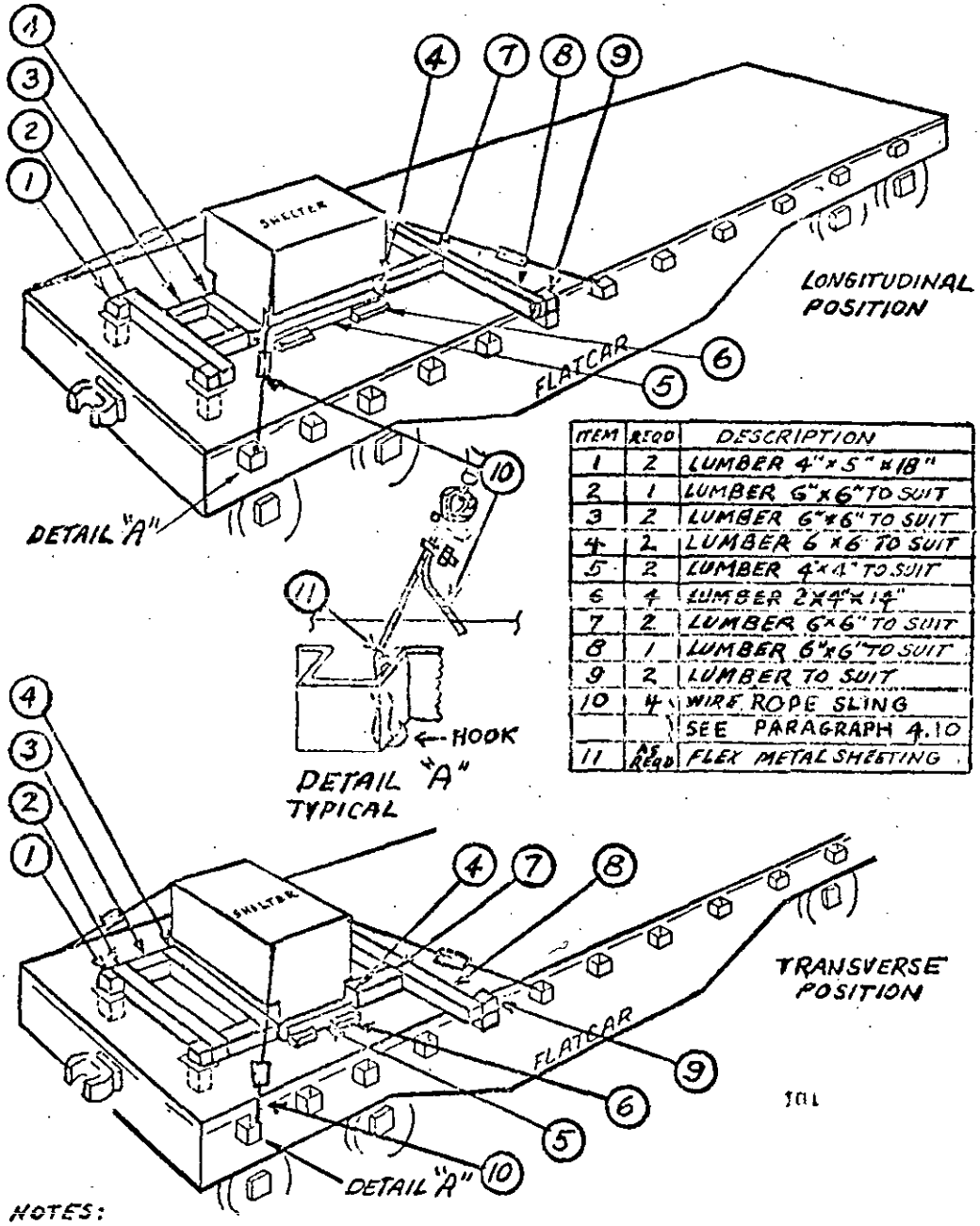
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4.17 Vehicular transportation test.- The shelter facility shall be secured and blocked to prevent shifting to a Government furnished 1 1/2 ton, M-715( ) truck and transported over the specially prepared course described in 4.17.1 for 10 laps at each of the speeds of 5 mph, 10 mph, 15 mph, and 18-20 mph. A lap is defined as traversing the course in one direction. The facility shall meet the requirements of paragraph 3.12 after completion of the test. (NOTE: The shelter facility shall be loaded to simulate the AN/GRC-122( ) using live and dummy loads (see 6.2d(2) and (3) for this test).

4.17.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch x 8-inch timbers placed 25 feet apart on the 8-inch face and with 4-inch face fully above ground. The top edges shall have a 1 x 1 inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be long enough to span the width of the vehicle (passing over them) and anchored down securely by any effective means. The surfaces on which the timbers are placed shall be hard and level.

4.18 Rail transport test.- The shelter facility, loaded to simulate the AN/GRC-122( ), using live and dummy loads (see 6.2d(2) and (3)), secured to a flatcar (the test flatcar) and coupled to two other flatcars, shall be impacted by a loaded gondola car. The test shall be run on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tiedown slings, (contractor furnished), shall be as specified in Figure 2 and shall meet the provisions of the test as specified in Paragraph 4.18.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in Paragraph 4.18.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9mph (or equivalent energy conditions acceptable to the Government Technical representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars, all with brakes off. All cars shall have standard (rigid) draft gear. A total of four (4) impacts shall be performed as follows: Two impacts with the shelter facility positioned longitudinally on the flatcar, one from each end of the shelter facility; two impacts with the shelter facility positioned transversely on the flatcar, one from each side of the shelter facility. The lashing, blocking, and wedges holding the shelter facility in place shall be tight at the at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact, the shelter facility shall meet the requirements of 3.13.

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ITEM	REQD	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 6"x6" TO SUIT
5	2	LUMBER 4"x4" TO SUIT
6	4	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	4	WIRE ROPE SLING SEE PARAGRAPH 4.10
11	AS REQD	FLEX METAL SHEETING

NOTES:
1- THE BLOCKING LUMBER SHALL BE NOTCHED TO ACCOMODATE TOWING EYES, BOLTS AND RIVET HEADS.
2- THE BLOCKING LUMBER SHALL BE RIGIDLY FASTENED TO THE FLOOR OF THE FLATCAR.

FIGURE 2

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4.18.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12-inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles; this shall be accomplished by means of sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall not be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 inch and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in figure referred to above. All other blocking shall be firmly oecured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter, 6 x 19, flexible steel, preformed-wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.18.2 Loaded gondola car.- The load in the loaded goudola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete, or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate form of the preceding acceptable materials, constrained to prevent shifting during impact.

4.19 Fungus test.- The equipment listed below shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810 to determine conformance with the requirements of 3.14. Spraying shall include parts and materials on and under the chassis or covers of equipments, and the interior of the equipment. The equipment shall remain in an open condition during the test. Operation is not required. The following equipments will be tested:

a. Power Signal Entrance Box

b. Power Distribution Panel SB-3358( )/GRC-142

c. Power Terminal Assembly

d. Switch Box SA-1650( )/GRC-142

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e. Interconnecting Boxes J-2728( )/GRC-142

f. Remote Box C-7279( )/GRC-142

g. Canvas parts

h. Tarpaulin

i. Chair w/seat belt

4.20 Salt fog test.- The equipment shall be tested in accordance with Method 509, Procedure I, of MIL-STD-810 to determine conformance with the requirements of 3.15. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test items by the test facility. The following pieces of equipment will be tested:

a. Switch Box SA-1650( )/GRC-142

b. Power Distribution Panel SB-3358( )/GRC-142

c. Interconnecting Boxes J-2728( )/GRC-142

d. Remote Box C-7279( )/GRC-142

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4.21 EMI test.- The shelter facility shall be tested in accordance with MIL-STD-462 with Notice 3 to determine conformance with the requirements of 3.16. The tests from MIL-STD-462 listed in the following paragraphs shall be performed using AC primary power (4.8.1k) and then repeated using DC primary power (4.8.1.j) except where specifically noted in the detailed test procedures.

a. Shelter, Electrical Equipment S-502( )/GRC-142( ):

CE01
CE02
CE04
RE02

b. Shelter, Electrical Equipment S-502( )/GRC-142( ) configured as the duplex Radio Teletypewriter Set AN/GRC-122( ) (These tests shall be run for information purposes only):

CE01	CE04	CS01	RE02
CE02	CE05	CS02	RE02.1
CE03	CE06	CS06	RS03*

*Method RS03 shall be run from 14 KHZ to 1 GHZ with the field intensity at 1 volt/meter and then rerun from 25 MHZ to 200 MHZ with the field intensity at 10 volts/meter.

4.22 Mounting insert tests.-

4.22.1 First Article insert test.- Four of each size insert (only inserts installed in mounting members shall be tested) shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table VIII. After test, the requirements of 3.17 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be performed prior to installation on any brackets, racks, etc.

4.22.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Only inserts installed in mounting members shall be tested. Each selected insert shall be tested in accordance with Table VIII. After test, the requirements of 3.17 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be performed prior to the installation of any brackets, racks, etc.

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TABLE VIII - Mounting Insert Loads

Insert Size	First Article		Production Test	
	Axial(lbs)	Torque(in-lbs)	Axial(lbs)	Torque(in-lbs)
5/16-18	2000	100	1000	100
1/4-20	600	50	375	50

4.23 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.28.4.

4.24 Inspection for dimensional interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.30. When a listed dimension is not within specified drawing or design limits it shall be considered a major defect.

a. Size and location of mounting positions for racks, operational equipment (including Government furnished equipment), electrical boxes and panels, and storage boxes and baskets.

b. Size and location of mounting positions on racks, operational equipment, electrical boxes and panels, and storage boxes and baskets.

c. Location and size of cavities in shelter walls.

d. External and internal dimensions of insertable assemblies mounted in or on shelter walls.

4.25 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, Table IX, and the requirements of 3.31.

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TABLE IX - Classification of Visual and Mechanical Defects

Classification	Defect
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3. Welds - cracked or porous. 4. Sealer improperly applied (holes, separations, or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire conduit not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc., inoperative. 9. Electrical receptacles, switches, jacks connectors, wire conduits or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Center pin of rf cable assemblies improperly crimped or soldered, loose, or protruding beyond outer edge of connector shell.
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges on bracket not removed. 5. Screws loose. 6. Gaskets, not completely and securely adhered.

4.26 Quality conformance inspection of packaging.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.27 Rough handling test (packaging).- When rough handling test in accordance with Specification Number MIL-P-115 is required by

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the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Inspection	Paragraph
Power Wiring	4.8.2
Signal Wiring	4.8.3
Visual and Mechanical	4.25

5. PACKAGING

5.1 Packaging requirements.- The packaging requirements shall be in accordance with MIL-S-55505.

6 NOTES

6.1 Intended use.- The Shelter Electrical Equipment S-502( )/GRC-142( ) is the shelter facility portion of the Radio Teletypewriter Set AN/GRC-142( ). Provisions are also provided in the shelter facility to convert the AN/GRC-142( ) to the duplex AN/GRC-122( ).

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

c. Level of Preservation and Packaging (See Section 5 above)

d. When GFE is furnished.

e. First Article inspection.

(1) Two sample units of each item cited in Section 1 are generally required so that lengthy testing can be divided between both units.

(2) Dummy loads (2 ea.) to simulate security equipment are required for certain tests. These dummy loads are considered test equipment. The dummy loads shall be 12 3/8 inches long, 14 inches wide, 11 1/4 inches high, and shall weigh 74 lbs. The center of gravity of the dummy loads shall be at their volumetric center. Provisions are necessary on the dummy loads to enable their mounting using the mounting hardware provided in the shelter facility for the security equipment.

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f. Submission of Statement of Treatment referenced in 3.26.

g. Technical literature required.

h. Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.26.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the con-xracting officer on the adequacy of corrective action (See 4.5.3.3).

6.5 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-142R. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this Specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Changes from previous issue.- Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-55347B (EL)

NAME OF ORGANIZATION AND ADDRESS
US Army CORADCOM
ATTN: DRDCO-CM-DM
Ft Monmouth, NJ 07703

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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